

Technology Stack

Date	26 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	2 Marks

Define Your Technology Stack:

The Flood Prediction and Early Warning System uses modern web and machine learning technologies. The selected technology stack ensures accurate predictions, easy maintenance, and future scalability.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	HTML, CSS, JavaScript	Provides a simple and responsive web interface for users to enter weather data and view flood prediction results.

2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python, Flask	Handles user requests, processes input data, loads the trained machine learning model, and generates predictions.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	CSV Dataset, Local File Storage	Stores historical weather data and trained machine learning model used for prediction.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	IBM Cloud	Enables scalable deployment and remote accessibility of the flood prediction <i>application</i> .

5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	GitHub	Manages source code versions and supports team collaboration during development.
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	Scikit-Learn, XGBoost, NumPy, Matplotlib, Jupyter Notebook	Used for data preprocessing, machine learning model training, evaluation, visualization, and development of the prediction system.